

EDUCATION University of Illinois Urbana-Champaign (UIUC) - *PhD Computer Science* 2020-2023
University of Washington - *MS Computational Linguistics* 2018-2020
Rensselaer Polytechnic Institute - *BS Computer Science* 2011-2014

EXPERIENCE *Founder & CEO - Zipf AI* 10/25-Present

- Built an agentic research platform from 0 to 1—multi-agent communication architectures, retrieval-augmented pipelines, and autonomous research workflows serving enterprise customers in finance, real estate, and biotech.
- Scaled to production: FastAPI backend, multi-provider LLM orchestration (Claude, Gemini, OpenAI), async distributed workers on SQS, Kubernetes (EKS) with KEDA autoscaling, PostgreSQL, Redis, and S3.
- Designed workflow engine for persistent web monitoring agents with semantic change detection, structured extraction, and signal scoring—enabling M&A signal detection, regulatory tracking, and competitive intelligence.

LLM & Agentic Strategy Consultant - AlphaSense, Wex Inc 08/23-08/24

- Advised on LLM strategy, agentic system design, and AI-powered research workflows for financial intelligence and enterprise platforms.

Senior Research Scientist, Tech Lead - Snowflake 05/23-10/25

- Led team of 7 researchers building retrieval, extraction, and generation systems for Snowflake Cortex Search. Shipped Arctic-Embed, a family of open-source embedding models achieving state-of-the-art retrieval accuracy with 20x lower inference costs than OpenAI and Cohere.
- Drove research partnerships with NVIDIA, Voyage, and Contextual AI for embedding models and RAG workflows, translating cutting-edge research into production systems serving enterprise customers.
- Built Snowflake’s research internship program from zero to 15 PhD interns and launched the Snowflake AI Fellowship with the University of Waterloo.

Senior Research Scientist - Neeva (Acquired by Snowflake) 12/22-05/23

- Designed and built a multi-phase LLM-powered retrieval pipeline from the ground up, implementing multi-vector retrieval, layout-aware representations, and model cascading.
- Compressed web summarization models from 7s to 300ms per batch via pruning, knowledge distillation, and inference optimization—enabling real-time search experiences.

Research Scientist Consultant - Neural Magic, Walmart, Qualtrics 10/20-12/22

- Developed and deployed production NLP systems for question answering, summarization, and retrieval. Applied model compression (quantization, pruning, distillation) achieving 4–40x throughput improvements without accuracy loss.

Applied Scientist / PM - Microsoft Research & AI, Bing 08/15-10/20

- Created the MS MARCO dataset and benchmark suite (3,500+ citations, 10,000+ researchers), establishing the standard for neural information retrieval and machine comprehension evaluation.
- Architected Bing’s real-time relevance metrics pipeline, scaling to \$7m/year across 16 markets with 1M+ weekly judgments. Developed novel stratified sampling methods using the Horvitz-Thompson estimator.

- Deployed Neural Machine Translation into Microsoft’s localization pipeline, saving \$3m/year through supplier contract renegotiation.

Research Leadership

Co-organizer of TREC Deep Learning Track (2018–2023), TREC RAG Track (2024–2026), and TREC Product Search Track (2023–2025)—defining community-wide benchmarks for retrieval, generation, and RAG system evaluation across 500+ research submissions.

Select Publications (7,000+ total citations, h-index 24)

- MS MARCO: A Human Generated MACHine Reading COMprehension Dataset - P. Bajaj, **D. Campos**, N. Craswell et al. – 3,500+ citations
- The Optimal BERT Surgeon: Scalable and Accurate Second-Order Pruning for Large Language Models - E. Kurtic, **D. Campos** et al. – EMNLP 2022, 216 citations
- Arctic-Embed: Scalable, Efficient, and Accurate Text Embedding Models - L. Merrick, D. Xu, G. Nuti, **D. Campos** – 2024
- Arctic-Embed 2.0: Multilingual Retrieval Without Compromise - P. Yu, L. Merrick, G. Nuti, **D. Campos** – 2024
- SuffixDecoding: A Model-Free Approach to Speeding Up Large Language Model Inference - G. Oliaro, Z. Jia, **D. Campos**, A. Qiao – NeurIPS 2024
- The Great Nugget Recall: Automating Fact Extraction and RAG Evaluation with LLMs - R. Pradeep, N. Thakur, S. Upadhyay, **D. Campos**, N. Craswell, J. Lin – SIGIR 2025
- STUN: Structured-Then-Unstructured Pruning for Scalable MoE Pruning - J. Lee, S. Hwang, A. Qiao, **D. Campos**, Z. Yao, Y. He – ACL 2025
- Inference Scaling for Bridging Retrieval and Augmented Generation - Y. Lee, S. Hwang, **D. Campos**, F. Gralinski, Z. Yao, Y. He – NAACL 2025
- Automating Generation of Long-Form Queries - S. Upadhyay, **D. Campos**, N. Thakur, R. Pradeep, N. Craswell, J. Lin – SIGIR 2026